

WASP-135b

R-1222

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_220305 · created 2026-07-31T08:54:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459643.9339 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.023
Transit depth (Rp/Rs)^2	1.93 +/- 0.64 [%]
Orbital Inclination (inc)	84.79 +/- 1.34
Airmass coefficient 1 (a1)	1.002 +/- 0.02
Airmass coefficient 2 (a2)	-0.0002 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.99 %
Best Comparison Star	#2 - [429, 291]
Optimal Aperture	2.66
Optimal Annulus	12.29
Transit Duration (day)	0.0825 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.139 $\pm$ 0.023 vs 0.139 (deviation 0.0 σ)
Duration fit vs published	0.0825 d vs 0.0687 d (deviation 1.84 σ)
Mid-time O-C	3.3 min
Depth SNR	4.6
Residual scatter	1.99 %

Light curve

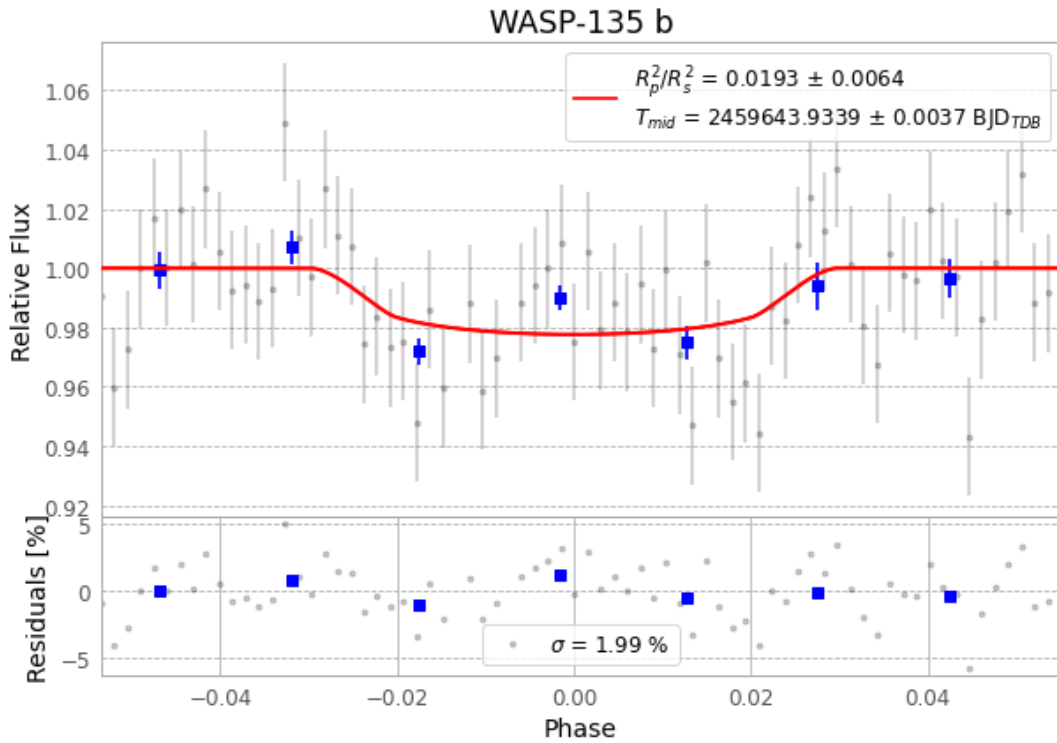


Figure 1 — FinalLightCurve_WASP-135 b_2022-03-05.png (SHA-256 f84605e7b2813f8f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 72fdd62b0f76af2d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-135220305083612.FITS → WASP-135220305121512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-220305.log (3dac8b834a20...), FinalLightCurve_WASP-135 b_2022-03-05.png (f84605e7b281...), FinalLightCurve_WASP-135 b_2022-03-05.csv (e4773be85e2f...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:54Z.